# Session 8 Lab Report

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**Course:** Android App Development using Gen AI
**Session:** 8 – UI Testing and Art Navigation
**Lab Title:** Console-Based Art Gallery Navigator

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## 1. Aim

To build a console-based Art Gallery Navigator in Kotlin using data classes and navigation functions.
The program allows users to move through a list of artworks using Next and Previous logic with wrapping functionality.

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## 2. Objective

- Practice Kotlin data classes
- Implement list management
- Apply conditional statements
- Implement wrapping navigation logic
- Improve modular programming skills

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## 3. Problem Statement

Create a console-based Art Gallery application that:

- Stores artwork details (title, artist, year)
- Displays artwork information in formatted output
- Navigates to the next artwork
- Navigates to the previous artwork
- Wraps to the first artwork if Next is pressed at the end
- Wraps to the last artwork if Previous is pressed at the beginning

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## 4. Approach

1. Created a data class `Artwork` to store artwork details.
2. Implemented `displayArtwork()` to print artwork details.

3. Implemented nextArtwork() using modular arithmetic for wrap logic.
4. Implemented previousArtwork() using conditional logic.
5. In the main function:
   - Created a list of artworks
   - Simulated Next and Previous navigation
   - Printed artwork details after each navigation

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# 5. Kotlin Program Code

```
data class Artwork(
   val title: String,
val artist: String,
val year: Int
)

fun displayArtwork(art: Artwork) {
   println("Viewing: ${art.title} (${art.artist}, ${art.year})")
println("-------------------------------------------") }

fun nextArtwork(currentIndex: Int, total: Int): Int {
return (currentIndex + 1) % total
}

fun previousArtwork(currentIndex: Int, total: Int): Int {
   return if (currentIndex - 1 < 0) {
      total - 1
} else {
      currentIndex - 1
   }
}

fun main() {

   val artworks = listOf(
      Artwork("Starry Night", "Vincent van Gogh", 1889),
      Artwork("Mona Lisa", "Leonardo da Vinci", 1503),
      Artwork("The Persistence of Memory", "Salvador Dali", 1931),
      Artwork("The Scream", "Edvard Munch", 1893)
   )

   var currentIndex = 0

   displayArtwork(artworks[currentIndex])

   println("-> Next Artwork")
   currentIndex = nextArtwork(currentIndex, artworks.size)
displayArtwork(artworks[currentIndex])

   println("-> Next Artwork")
```

```
    currentIndex = nextArtwork(currentIndex, artworks.size)
displayArtwork(artworks[currentIndex])

    println("-> Next Artwork")
    currentIndex = nextArtwork(currentIndex, artworks.size)
    displayArtwork(artworks[currentIndex])

    println("-> Previous Artwork")
    currentIndex = previousArtwork(currentIndex, artworks.size)
displayArtwork(artworks[currentIndex])
}
```

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# 6. Sample Output

```
Viewing: Starry Night (Vincent van Gogh, 1889)
-> Next Artwork
Viewing: Mona Lisa (Leonardo da Vinci, 1503)
-> Next Artwork
Viewing: The Persistence of Memory (Salvador Dali, 1931)
-> Next Artwork
Viewing: The Scream (Edvard Munch, 1893)
-> Previous Artwork
Viewing: The Persistence of Memory (Salvador Dali, 1931)
```

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# 7. Key Concepts Used

- Data Class in Kotlin
- List Collection
- Modular Arithmetic
- Conditional Statements
- Index Wrapping Logic
- Function Modularity

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# 8. Learning Outcomes

Through this lab, I learned:

- How to structure Kotlin programs using data classes
- How navigation logic works using index manipulation
- How modular arithmetic simplifies wrap-around behaviour
- How to separate display logic from navigation logic
- Importance of clean and reusable functions

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# 9. Conclusion

The console-based Art Gallery Navigator was successfully implemented.
The application correctly displays artworks and supports Next and Previous navigation with proper wrapping logic.
This lab strengthened my understanding of Kotlin fundamentals and prepared me for implementing similar navigation logic in Jetpack Compose UI applications.